



ABV-Indian Institute of Information Technology & Management Gwalior
Mid Semester Examination

Course Name: IoT and Big Data Management (MS-607)

Date of Examination: 23 September 2024 (4:00-6:00p.m.)

Note: Attempt all questions

Maximum Marks: 30

Duration: 2 hours

1. What is Big Data. What are the benefits of using big data analytics in business? Elaborate the discussion with relevant case study. (6)

2. Describe with the aid of a diagram, the Hadoop distributed file system architecture, and explain how it stores data across a cluster of machines. (6)

3. a) Show MapReduce process diagrammatically to depict a client submitting a job, the workflow of JobTracker and TaskTracker. (3)

b) How does the data convert to (key, value) pairs before passing to the Mapper? How do the InputSplit and RecordReader function? (3)

4. Describe the architecture of Hadoop YARN. A large e-commerce company has started using Hadoop YARN for processing big data. They notice that some jobs are taking too long to complete, and the system seems to have idle resources. How would you approach optimizing their YARN resource utilization? (6)

5. Give an overview on:

- a) Zookeeper b) Apache Spark c) Apache Mahout d) Pig e) Hive

(6)

(6)

over
exam
after



ABV-Indian Institute of Information Technology & Management Gwalior
End Semester Examination

Course Name: IoT and Big Data Management (MS-607)

Date of Examination: 30 December 2024 (10:00 - 1:00p.m.)

Note: Attempt all questions

Maximum Marks: 50
hours

Duration: 3

1. Explain the core components of the Hadoop ecosystem and elaborate on their roles and functionalities in detail. (10)
2. How do ACID and BASE properties differ? Why is the consistency not enforceable in NoSQL distributed databases during a transaction processing? Elaborate the discussion with example. (10)
3. Data architecture pattern can be selected from the following architectures: relational SQL table, key-value pair, column-family, document and graph DBs. Explain with an example, *how and when each of these is used*. (10)
4. What are the features present in the Apache Spark architecture that enable fast computations. Describe the function of Spark SQL, Spark Streaming, MLlib and GraphX? (10)
5. Discuss the architecture of Apache HBASE. Also, discuss its read and write operation. (5)
6. Discuss the application of IoT technology in healthcare sector. Elaborate your discussion with use cases. (5)



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ABV-Indian Institute of Information Technology and Management Gwalior

Minor Examination, Academic Year 2024-25

MBA-I Semester

Course Name: IoT and Big Data Management

Date of Examination: 29th November 2025

Timing: 02:00-05:00 P.M

Maximum Marks: 40

Duration: 3 hours

1. A car company records weekly sales for each showroom. A sample of the weekly sales data is given below:

Showroom_ID	Week_Number	Jaguar_Land_Rover	Hexa	Zest	Nexon	Safari
SR001	1	4	6	2	7	5
SR002	1	3	5	4	6	4
SR003	1	5	4	3	8	6

- (a) Compute the total weekly sales for each showroom and represent the result in a key-value pair format.
- (b) Represent the weekly sales record of Showroom SR001 in a JSON document format as it would be stored in a document-oriented database. (6)

2. Explain Apache Spark. Describe its key features such as in-memory processing, fault tolerance, scalability, and support for multiple programming APIs. Why is Spark preferred over traditional MapReduce for big data processing? (6)

3. How does the Name node determine which data node to write to? Explain the data writing procedure in HDFS with the help of a diagram. (6)

4. Explain the architecture of Apache HBase. Describe the roles of HMaster, RegionServer, Regions, and Zookeeper in managing distributed storage. (6)

5. Write a short note on: MapReduce, Mahout, Pig, Flume, and Sqoop. (6)

6. Differentiate between SQL and NoSQL Datastore ACID and BASE properties HBASE and HDFS (6)

7. Big Data is transforming modern organizations in their strategic, operational, and customer-related decisions. Discuss any three major business applications of Big Data and provide suitable examples to support your answer. (4)



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ABV-Indian Institute of Information Technology and Management Gwalior

Minor Examination, Academic Year 2024-25
Course Name: IoT and Big Data Management
MBA-I Semester
Date of Examination: 23rd September 2025
Timing: 04:00-06:00 P.M

Note: (1) Attempt all questions
(2) Each question carries equal marks

Maximum Marks: 30

Duration: 2 hours

1. How do the CISCO reference model and Oracle reference architecture correlate in an IoT architecture. (6)
2. How can an IoT-based reference model be used to design a smart parking architecture, and what would be the key layers, components, and technologies involved in this architecture? (6)
3. Explain the architecture of Hadoop distributed file system (HDFS) with the role of NameNode and DataNodes. How does HDFS ensure fault tolerance? (6)
4. Suppose a client writes a 300 MB file into HDFS with a block size of 128 MB and a replication factor of 3. Explain how many blocks will be created, and how the write and subsequent read operations will take place. (6)
5. (a) Describe how ResourceManager, NodeManager, and ApplicationMaster interact when a job is submitted. (3)
(b) Describe the end-to-end flow of a MapReduce job, highlighting the roles of InputSplit, RecordReader, Mapper, Shuffle/Sort, Combiner, and Reducer. (3)

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6. Differentiate between (6)
- SQL and NoSQL Datastore
ACID and BASE properties
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7. Big Data is transforming modern organizations in their strategic, operational, and customer-related decisions. Discuss any three major business applications of Big Data and provide suitable examples to support your answer. (4)